

Yehyun Suh

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EDUCATION

Vanderbilt University

Doctor of Philosophy in Computer Science

Present
Nashville, TN

Dongguk University

Bachelor of Science in Information & Communication Engineering

Feb 2022
Seoul, South Korea

WORKING EXPERIENCE

AlphaTEC Spine

Research Intern (Principal Investigator: Kongbin Kang, Ph.D.)

June 2024 – Aug 2024
Carlsbad, CA

Subdream Studios

Game Developer Intern

June 2021 – Aug 2021
Remote

Innovation Center for Engineering Education

Working Scholar

Jan 2021 – Dec 2021
Seoul, South Korea

College of Engineering Office

Working Scholar

Mar 2020 – Jan 2021
Seoul, South Korea

RESEARCH EXPERIENCE

Visual Information Neuroimaging Engineering (VINE) Lab, Vanderbilt University

Graduate Student Researcher (Advisor: Daniel Moyer, Ph.D.)

Present
Nashville, TN

innovative Radiology AI Laboratory (iRAIL), Seoul National University Hospital

Research Intern (Principal Investigator: Jisun Hwang, M.D.)

May 2022 – Aug 2022
Seoul, South Korea

Intelligent Software Architecture (ISA) Lab, Dongguk University

Undergraduate Student Researcher (Advisor: Woongsup Kim, Ph.D.)

Mar 2020 – Feb 2022
Seoul, South Korea

PUBLICATIONS

* denotes equal contribution

Under Review

[U3] **Differentiable Projective Ellipse Registration for Single-View Fluoroscopic Pose Estimation of Acetabular Implants**

Yehyun Suh, Emily R. Oleisky, Hao Yang, Jie Ying Wu, J. Ryan Martin, Daniel Moyer

[U2] **Linking Deformation-Derived Spectral Embeddings to Representative Pelvic Anatomy**

Yehyun Suh, J. Ryan Martin, Daniel Moyer

[U1] **Reducing Parallax Error in Intraoperative Fluoroscopy: A Perspective-Projection Method for Accurate Cup Orientation in Total Hip Arthroplasty**

Emily R. Oleisky*, Yehyun Suh*, Daniel Faraj, Zayn Kayali, Paul Solimon, Wes Mayes, Courtney Baker, Daniel Moyer, J. Ryan Martin

Journal

[J4] **Development of a Machine Learning Model for Determining Alignment in Knees Following Total Knee Arthroplasty**

Anoop Chandrashekar*, Yehyun Suh*, Jacob Fox, Aleksander Mika, Daniel Moyer, Gregory Polkoswski, Martin Faschingbauer, J. Ryan Martin

Journal of Arthroplasty (IF: 3.4, JCR Q1, Top 10.7 %), 2025

- [J3] **Development of a Deep Learning Model for Automating Implant Position in Total Hip Arthroplasty**
Peter Chan, Courtney E. Baker, **Yehyun Suh**, Daniel Moyer, J. Ryan Martin
Journal of Arthroplasty (IF: 3.4, JCR Q1, Top 10.7 %), 2025
- [J2] **Novel Dilation-Erosion Labeling Technique Allows for Rapid, Accurate and Adjustable Alignment Measurements in Primary Total Knee**
Aleksander Mika, **Yehyun Suh**, Robert Elrod, Daniel Moyer, Martin Faschingbauer, J. Ryan Martin
Computers in Biology and Medicine (IF: 7.0, JCR Q1, Top 6.0 %), 2025
- [J1] **Deep Learning-based Fully Automatic Risser Stage Assessment Model using Abdominal Radiographs**
Jae-Yeon Hwang, Yisak Kim, Jisun Hwang, **Yehyun Suh**, Sook Min Hwang, Hyeyun Lee, Minsu Park
Pediatric Radiology (IF: 2.1, JCR Q2, Top 37.5 %), 2024

Conference

- [C5] **Landmark Detection Uncertainty as a Reliability Weight for Robust Landmark-based 2D/3D Pelvic Pose Estimation**
Yehyun Suh, Brayden Schott, Chuo Mo, J. Ryan Martin, Daniel Moyer
Medical Imaging with Deep Learning (MIDL), 2026, Taipei, Taiwan, ROC
- [C4] **Predicting 2D and 3D Leg Motion from Short-Leg Films in Total Knee Arthroplasty**
Zheyue Caroline Yao, **Yehyun Suh**, Anoop S. Chandrashekar, J. Ryan Martin, Daniel Moyer
SPIE Medical Imaging, 2026, Vancouver, Canada
- [C3] Oral Presentation
Enhanced Landmark Detection Model in Pelvic Fluoroscopy using 2D/3D Registration Loss
Chou Mo*, **Yehyun Suh***, J. Ryan Martin, Daniel Moyer
SPIE Medical Imaging, 2026, Vancouver, Canada
- [C2] **Pedicle Screw Pairing and Registration for Screw Pose Estimation from dual C-arm Images Using CAD Models**
Yehyun Suh, Lin Li, Aric Plumley, Chaochao Zhou, Daniel Moyer, Kongbin Kang
SPIE Medical Imaging, 2026, Vancouver, Canada
- [C1] Oral Presentation
Landmark Prediction in Large Radiographs using RoI based Label Augmentation
Mahd Mohsin*, **Yehyun Suh***, Anoop Chandrashekar, J. Ryan Martin, Daniel Moyer
SPIE Medical Imaging, 2025, San Deigo, California

Workshop / Short Paper

- [W4] Oral Presentation
2D/3D Registration of Acetabular Hip Implants Under Perspective Projection and Fully Differentiable Ellipse Fitting
Yehyun Suh, J. Ryan Martin, Daniel Moyer
MICCAI Workshop on Clinical Image-based Procedures (CLIP), 2025, Daejeon, South Korea
- [W3] **Introducing Beyond FA: Crowdsourcing Imaging Biomarkers for Alzheimer's Disease**
Elyssa McMaster, Nancy Newlin, Trent Schwartz, Adam Saunders, Gaurav Rudravaram, **Yehyun Suh**, Jongyeon Yoon, Michael Kim, Chloe Cho, Karthik Ramadass, Yihao Liu, Lianrui Zuo, Eleftherios Garyfallidis, Talia Nir, Neda Jahanshad, Kurt Schilling, Daniel Moyer, Bennett Landman
Medical Imaging with Deep Learning (MIDL), short paper track, 2025, Salt Lake City, Utah
- [W2] **Label Augmentation Method for Medical Landmark Detection in Hip Radiograph Images**
Yehyun Suh, Peter Chan, J. Ryan Martin, Daniel Moyer

NeurIPS workshop on Medical Imaging (MedNeurIPS), 2023, New Orleans, Louisiana

- [W1] **Dilation-Erosion Methods for Radiograph Annotation in Total Knee Replacement**
Yehyun Suh, Aleksander Mika, J. Ryan Martin, Daniel Moyer
Medical Imaging with Deep Learning (MIDL), short paper track, 2023, Nashville, Tennessee

Intellectual Property

- [I2] **2D/3D Registration of Acetabular Hip Implants Under Perspective Projection and Fully Differentiable Ellipse Fitting**
Yehyun Suh, J. Ryan Martin, Daniel Moyer
- [I1] **Pedicle Screw Orientation Estimation Algorithm**
Kongbin Kang, **Yehyun Suh**, Lin Li, Chaochao Zhou, Aric Plumley, Robert Winn

Organized Challenge

- [O1] **Beyond FA**
Elyssa McMaster, Nancy Newlin, Chloe Cho, Gaurav Rudravaram, Karthik Ramadass, Adam Saunders, Jongyeon Yoon, **Yehyun Suh**, Trent Schwartz, Michael Kim, Yihao Liu, Lianrui Zuo, Kurt Schilling, Talia Nir, Neda Jahanshad, Daniel Moyer, Eleftherios Garyfallidis, Bennett Landman

Preprint

- [P1] **Better Pose Initialization for Fast and Robust 2D/3D Pelvis Registration**
Yehyun Suh, J. Ryan Martin, Daniel Moyer

Abstract

- [A2] **Pose Estimation of Acetabular Hip Implants in Total Hip Arthroplasty Under Perspective Projection**
Yehyun Suh, J. Ryan Martin, Daniel Moyer
13th Annual Surgery, Intervention, and Engineering Symposium, 2024, Nashville, Tennessee
- [A1] Best Poster Award
Label Augmentation Method for Medical Landmark Detection in Hip Radiograph Images
Yehyun Suh, Peter Chan, Aleksander Mika, J. Ryan Martin, Daniel Moyer
12th Annual Surgery, Intervention, and Engineering Symposium, 2023, Nashville, Tennessee

Korean Book [Participated as an undergraduate research assistant]

- [KB3] 김웅섭, Software engineering - Python, Flask, Heroku, 2021, doi.org/10.979.1191057/478
- [KB2] 김웅섭, 소고기보다 알고리즘, 2021, doi.org/10.979.1191057/850
- [KB1] 김웅섭, Mobile software - HTML, JavaScript, React, 2020

HONORS AND AWARDS

Graduate

- **NIH Predoctoral Individual NRSA F31 Fellowship**, NIBIB, NIH, 2026
- **Best Poster Award**, 12th Annual Surgery, Intervention, and Engineering Symposium, 2023

Undergraduate

- **Summa Cum Laude**, 2022, Dongguk University
- Dean's List, Dongguk University, 2016 Spring/2016 Fall/2020 Spring/2020 Fall/2021 Spring
- **Software Academic Excellence Scholarship**, Dongguk University, 2020 Fall/2021 Spring
- **Academic Excellence Scholarship**, Dongguk University, 2016 Spring/2016 Fall/2021 Spring
- 3rd Place in XR Challenge Competition, Ministry of Science and ICT, Korea, 2021
- 3rd Place in Summer Dongguk Capstone Value Up Program, Dongguk University, 2021

- 2nd Place in Graduation Project Competition, Dongguk University, 2021
- 4th Place in Winter Dongguk Capstone Value Up Program, Dongguk University, 2021
- Course Excellence Scholarship, Enterprise & Society Tailored Capstone Design, Dongguk University, 2021
- Course Excellence Scholarship, Operating System, Dongguk University, 2020
- Oversea Academic Tour Scholarship, Dongguk University, 2017

TEACHING EXPERIENCE

DS 5760 No SQL for Modern Data Science Applications , Vanderbilt University <i>Graduate Teaching Assistant</i> (Instructor: Peng (Dana) Zhang, Ph.D.)	2025 Fall Nashville, TN
DS 5460 Big Data Scaling , Vanderbilt University <i>Graduate Teaching Assistant</i> (Instructor: Peng (Dana) Zhang, Ph.D.)	2025 Spring Nashville, TN
CS 6357 Open-Source Programming for Medical Image Analysis , Vanderbilt University <i>Graduate Teaching Assistant</i> (Instructor: Ipek Oguz, Ph.D.)	2024 Fall Nashville, TN
CS 6357 Open-Source Programming for Medical Image Analysis , Vanderbilt University <i>Graduate Teaching Assistant</i> (Instructor: Ipek Oguz, Ph.D.)	2023 Fall Nashville, TN
CS 1101 Programming and Problem Solving , Vanderbilt University <i>Graduate Teaching Assistant</i> (Instructor: Rui (Gina) Bai, Ph.D.)	2023 Spring Nashville, TN
RGC 1046 Artificial Intelligence in the Future Society , Dongguk University <i>Undergraduate Teaching Assistant</i> (Instructor: Sungyoun Choi, Ph.D.)	2021 Fall Seoul, South Korea
CSE 2025 Computational Thinking , Dongguk University <i>Undergraduate Teaching Assistant</i> (Instructor: Sungyoun Choi, Ph.D.)	2021 Spring Seoul, South Korea

MENTORING EXPERIENCE

Zheye (Caroline) Yao , Undergraduate Student at Swarthmore College <i>Femoral Motion Estimation</i> - Publications: [C4] - Next Position: Ph.D. Student at University of Chicago	May 2025 – Aug 2025
Chou (Athena) Mo , Undergraduate Student at UCLA <i>Pose Estimation Error based Landmark Detection</i> - Publications: [C3] - Next Position: Undergraduate Researcher at UCLA	May 2025 – Aug 2025
Yuyang Xing , Undergraduate Student at Vanderbilt University <i>3D Pelvis Segmentation</i> - Next Position: M.S. Student at University of Texas at Austin	May 2025 – Aug 2025
Prashan Sapkota , Undergraduate Student at Fisk University <i>Nano Particle Segmentation</i>	May 2025 – Aug 2025
Mahd Mohsin , Undergraduate Student at Vanderbilt University <i>Landmark Prediction on Large Scale Radiograph Images</i> - Publications: [C1] - Next Position: Ph.D. Student at University of North Carolina at Chapel Hill	Jan 2024 – Feb 2025
Cheng Qiu , Undergraduate Student at Vanderbilt University <i>Hybrid Graph U-Net for Image Segmentation</i> - Next Position: Ph.D. Student at Stony Brook University	May 2023 – Aug 2023

INVITED TALKS & TUTORIALS

Diagnostics and Dashboard	Aug 2025
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Vanderbilt University	Nashville, TN
- Link: https://www.vanderbilt.edu/valiant/ai-summer-school-2025/	
Diagnostics and Dashboard	Aug 2024
Vanderbilt University	Nashville, TN
- Link: https://www.vanderbilt.edu/valiant/ai-summer-school/	
Performance Analysis and Output Visualization in Medical Imaging	Oct 2024
Vanderbilt University	Nashville, TN
Artificial Intelligence in Medical Scenes	Oct 2023
Tennessee State University	Nashville, TN
- Link: https://www.linkedin.com/posts/vu-electrical-and-computer-engineering_we-had-a-wonderful-kickoff-event-for-our-activity-7122657323675746304-0_cG/	
Implementing Data Science Projects in Python	Aug 2023
Vanderbilt University	Nashville, TN
- Link: https://www.linkedin.com/feed/update/urn:li:activity:7097981972118994944/	

PROJECTS

Reverse Pill Image Search , Boostcamp AI Tech	Apr 2022 – June 2022
- People who were excluded from medical benefits did not receive an appropriate prescription in time - Developed a model that can search the most similar pill from a single pill image given from the user by extracting features such as shape, type, and color of pills through image classification - Final performance of the model - top1 accuracy: 43%, top5 accuracy: 80% (https://github.com/boostcampaitech3/final-project-level3-cv-16)	

EXTRACURRICULAR ACTIVITIES

Vanderbilt Lab for Immersive AI Translation (VALIANT) , Vanderbilt University	Present
<i>Board Member / Treasurer</i>	Nashville, TN
Korean Graduate Students and Scholars Association (KSSA) , Vanderbilt University	Aug 2022 – Aug 2023
<i>Secretary</i>	Nashville, TN
Boostcamp AI Tech , Naver Connect Foundation	Jan 2022 – June 2022
<i>Computer Vision Track</i>	Seoul, South Korea
Innovation Center for Engineering Education , Dongguk University	2021
<i>President of the Student Committee 'DAVINCI'</i>	Seoul, South Korea
Information & Communication Engineering , Dongguk University	2020
<i>Vice President of the Student Committee</i>	Seoul, South Korea
Dongguk University Tennis Club	Sep 2016 – Feb 2022
<i>Regular Member / Board Member / Vice President</i>	Seoul, South Korea

REFERENCES

Daniel Moyer, Ph.D., Assistant Professor
Department of Computer Science
School of Engineering
Vanderbilt University
daniel.moyer@vanderbilt.edu

J. Ryan Martin, M.D., Associate Professor
Department of Orthopaedic Surgery
Vanderbilt University Medical Center
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